General Bite #5: Components as RF Parts -- Passives and ICs

Capacitors, resistors, inductors, diodes, batteries, vacuum tubes; integrated circuits, MMICs, op-amps, RF connectors, ferrite cores.

Groups G6A + G6B · 23 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

Part 1: Passive Components And Discretes (G6A)

Batteries

Question	Answer	Q
Minimum discharge voltage for max life of a 12 V lead-acid battery	**10.5 V** (G6A01).	G6A01
Advantage of low-internal-resistance batteries	**High discharge current** (G6A02). Important for transmitter peak draw.	G6A02

Diode Forward Voltage

Question	Answer	Q
Germanium diode forward threshold	**0.3 V** (G6A03). Used in low-signal RF detectors because the small drop preserves weak signals.	G6A03
Silicon diode forward threshold	**0.7 V** (G6A05). The standard PN junction drop.	G6A05

Capacitors

Question	Answer	Q
Characteristic of an electrolytic capacitor	**High capacitance for a given volume** (G6A04). Tradeoff: polarity matters, leaky, not great at RF.	G6A04
Characteristic of low-voltage ceramic capacitors	**Comparatively low cost** (G6A08).	G6A08

Resistors And Inductors At RF

Question	Answer	Q
Why no wire-wound resistors in RF circuits?	**The resistor's inductance could make circuit performance unpredictable** (G6A06). Wire-wound = coil + resistance.	G6A06
What happens above an inductor's self-resonant frequency	**It becomes capacitive** (G6A11). Stray inter-turn capacitance starts dominating.	G6A11

Transistors And MOSFETs

Question	Answer	Q
Bipolar transistor used as a switch operates between…	**Saturation and cutoff** (G6A07). Either full-on or full-off.	G6A07
MOSFET construction	**The gate is separated from the channel by a thin insulating layer** (G6A09). The "MOS" = metal-oxide-semiconductor sandwich.	G6A09

Vacuum Tubes

Question	Answer	Q
Which element regulates electron flow between cathode and plate?	**The control grid** (G6A10). Bias on the grid controls plate current.	G6A10
Primary purpose of the screen grid	**To reduce grid-to-plate capacitance** (G6A12). Without it, signal coupling between input and output limits gain at HF.	G6A12

Part 2: Integrated Circuits And RF Hardware (G6B)

Ferrite Cores

Question	Answer	Q
What determines ferrite-core performance at different frequencies?	**The composition (or "mix") of materials used** (G6B01). Different mixes target HF, VHF, or VHF/UHF.	G6B01
Advantage of a ferrite-core toroidal inductor	Large L values + mix optimized for frequency + field contained in core – **all of the above** (G6B05).	G6B05
How a ferrite bead reduces common-mode RF current on coax shield	**By creating an impedance in the current's path** (G6B10). Adds lossy impedance only at RF.	G6B10

Integrated Circuits

Question	Answer	Q
What does MMIC stand for?	**Monolithic Microwave Integrated Circuit** (G6B02).	G6B02

CMOS advantage vs TTL	**Low power consumption** (G6B03).	G6B03
Op-amp is what type of device?	**Analog** (G6B06). Continuous voltages in, continuous voltage out.	G6B06
How is an LED biased when emitting light?	**Forward biased** (G6B08).	G6B08

RF Connectors

Question	Answer	Q
Upper frequency limit for low-SWR BNC operation	**4 GHz** (G6B04). BNC is good through microwave; PL-259 is HF only.	G6B04
Type N connector	**A moisture-resistant RF connector useful to 10 GHz** (G6B07). Threaded, weatherproof.	G6B07
What is an SMA connector?	**A small threaded connector suitable for signals up to several GHz** (G6B11).	G6B11
Connector for low-frequency or DC signal connections to a transceiver	**RCA Phono** (G6B12). The classic audio/keyer/control-line connector.	G6B12

Mental Cheat Sheet

- 12 V lead-acid min discharge: **10.5 V**.
- Low-R battery → high discharge current.
- Germanium diode V_F ≈ **0.3 V**; silicon ≈ **0.7 V**.
- Electrolytic cap = high C per volume, polarized.
- Ceramic low-V cap = cheap.
- No wire-wound R at RF → unwanted **inductance**.
- Above self-resonance an inductor looks **capacitive**.
- BJT switch: **saturation and cutoff**.
- MOSFET gate: **insulated** from channel by thin oxide.
- Vacuum tube control grid = electron-flow regulator.
- Screen grid reduces **grid-plate capacitance**.
- Ferrite-core behavior set by **mix**.
- Ferrite bead adds **impedance** to common-mode RF.
- MMIC = Monolithic Microwave IC.
- CMOS advantage = **low power**.
- Op-amp = **analog** device.
- LED **forward** biased to emit.
- BNC works to **4 GHz**; type N to **10 GHz**; SMA **several GHz**.
- RCA phono = low-freq / DC interconnect.

Common Traps

- Lead-acid min is **10.5 V**, not 8.5 or 12.
- Silicon V_F is **0.7 V**; germanium is **0.3 V**. Don't swap.
- Electrolytic = **high** C per volume, NOT tight tolerance / low leakage.
- Wire-wound R is bad at RF because of **inductance**, not capacitance.
- Above self-resonance an inductor goes **capacitive**, not "harmonics."
- MOSFET has **insulated** gate (oxide layer), not back-biased junction.
- Screen grid reduces **grid-to-plate capacitance**, not "plate resistance."
- MMIC = **Monolithic Microwave**, not "multi-mode" or "metal-monolayer."
- Op-amp is **analog**, not digital, not programmable logic.
- LED is **forward** biased, not reverse or Zener.
- SMA = **threaded** connector; BNC = **bayonet**; PL-259 = HF screw-on.

Takeaway: Passive parts are not ideal at RF: wire-wound R has unwanted L, electrolytic C is polarized and poor at high frequency, and inductors above self-resonance look capacitive. Silicon junction drops 0.7 V; germanium drops 0.3 V; LED is forward-biased. Ferrite cores work because of the **mix**; ferrite beads add impedance to suppress common-mode RF. MMIC = monolithic microwave IC; op-amp is analog. Connector frequency limits: BNC 4 GHz, N 10 GHz, SMA several GHz, RCA for DC/audio.

Bite 5 Question Bank (G6A + G6B)

G6A01. What is the minimum allowable discharge voltage for maximum life of a standard 12-volt lead-acid battery?
A. 6 volts.
B. 8.5 volts.
C. 10.5 volts.
D. 12 volts.

G6A02. What is an advantage of batteries with low internal resistance?
A. Long life.
B. High discharge current.
C. High voltage.
D. Rapid recharge.

G6A03. What is the approximate forward threshold voltage of a germanium diode?
A. 0.1 volt.
B. 0.3 volts.
C. 0.7 volts.
D. 1.0 volts.

G6A04. Which of the following is characteristic of an electrolytic capacitor?
A. Tight tolerance.
B. Much less leakage than any other type.
C. High capacitance for a given volume.
D. Inexpensive RF capacitor.

G6A05. What is the approximate forward threshold voltage of a silicon junction diode?
A. 0.1 volt.
B. 0.3 volts.
C. 0.7 volts.
D. 1.0 volts.

G6A06. Why should wire-wound resistors not be used in RF circuits?
A. The resistor's tolerance value would not be adequate.
B. The resistor's inductance could make circuit performance unpredictable.
C. The resistor could overheat.
D. The resistor's internal capacitance would detune the circuit.

G6A07. What are the operating points for a bipolar transistor used as a switch?
A. Saturation and cutoff.
B. The active region (between cutoff and saturation).
C. Peak and valley current points.
D. Enhancement and depletion modes.

G6A08. Which of the following is characteristic of low voltage ceramic capacitors?
A. Tight tolerance.
B. High stability.
C. High capacitance for given volume.
D. Comparatively low cost.

G6A09. Which of the following describes MOSFET construction?
A. The gate is formed by a back-biased junction.
B. The gate is separated from the channel by a thin insulating layer.
C. The source is separated from the drain by a thin insulating layer.
D. The source is formed by depositing metal on silicon.

G6A10. Which element of a vacuum tube regulates the flow of electrons between cathode and plate?
A. Control grid.
B. Suppressor grid.
C. Screen grid.
D. Trigger electrode.

G6A11. What happens when an inductor is operated above its self-resonant frequency?
A. Its reactance increases.
B. Harmonics are generated.
C. It becomes capacitive.
D. Catastrophic failure is likely.

G6A12. What is the primary purpose of a screen grid in a vacuum tube?
A. To reduce grid-to-plate capacitance.
B. To increase efficiency.
C. To increase the control grid resistance.
D. To decrease plate resistance.

G6B01. What determines the performance of a ferrite core at different frequencies?
A. Its conductivity.
B. Its thickness.
C. The composition, or “mix,” of materials used.
D. The ratio of outer diameter to inner diameter.

G6B02. What is meant by the term MMIC?
A. Multi-Mode Integrated Circuit.
B. Monolithic Microwave Integrated Circuit.
C. Metal Monolayer Integrated Circuit.
D. Mode Modulated Integrated Circuit.

G6B03. Which of the following is an advantage of CMOS integrated circuits compared to TTL integrated circuits?
A. Low power consumption.
B. High power handling capability.
C. Better suited for RF amplification.
D. Better suited for power supply regulation.

G6B04. What is a typical upper frequency limit for low SWR operation of 50-ohm BNC connectors?
A. 50 MHz.
B. 500 MHz.
C. 4 GHz.
D. 40 GHz.

G6B05. What is an advantage of using a ferrite core toroidal inductor?
A. Large values of inductance may be obtained.
B. The magnetic properties of the core may be optimized for a specific range of frequencies.
C. Most of the magnetic field is contained in the core.
D. All these choices are correct.

G6B06. What kind of device is an integrated circuit operational amplifier?
A. Digital.
B. MMIC.
C. Programmable Logic.
D. Analog.

G6B07. Which of the following describes a type N connector?
A. A moisture-resistant RF connector useful to 10 GHz.
B. A small bayonet connector used for data circuits.
C. A low noise figure VHF connector.
D. A nickel plated version of the PL-259.

G6B08. How is an LED biased when emitting light?
A. In the tunnel-effect region.
B. At the Zener voltage.
C. Reverse biased.
D. Forward biased.

G6B10. How does a ferrite bead or core reduce common-mode RF current on the shield of a coaxial cable?
A. By creating an impedance in the current’s path.
B. It converts common-mode current to differential mode current.
C. By creating an out-of-phase current to cancel the common-mode current.
D. Ferrites expel magnetic fields.

G6B11. What is an SMA connector?
A. A type-S to type-M adaptor.
B. A small threaded connector suitable for signals up to several GHz.
C. A connector designed for serial multiple access signals.
D. A type of push-on connector intended for high-voltage applications.

G6B12. Which of these connector types is commonly used for low frequency or dc signal connections to a transceiver?
A. PL-259.
B. BNC.
C. RCA Phono.
D. Type N.

Answer Key

G6A01: C	G6A02: B	G6A03: B	G6A04: C	G6A05: C
G6A06: B	G6A07: A	G6A08: D	G6A09: B	G6A10: A
G6A11: C	G6A12: A	G6B01: C	G6B02: B	G6B03: A
G6B04: C	G6B05: D	G6B06: D	G6B07: A	G6B08: D
G6B10: A	G6B11: B	G6B12: C		